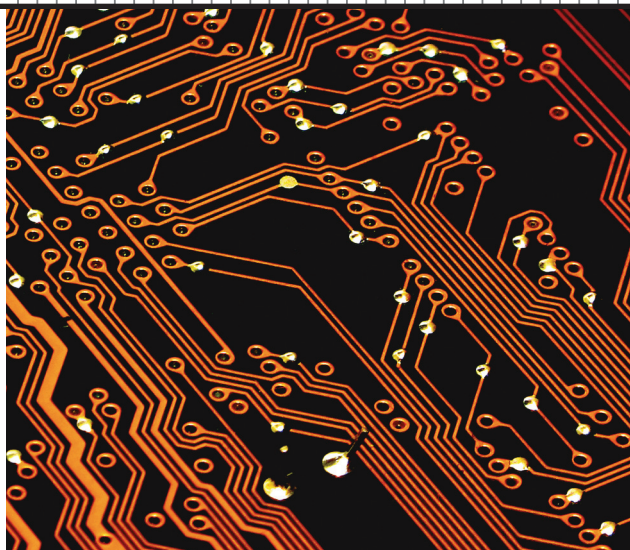


The Three Rs of Cyberphysical Spaces



Vivek Menon, *Amrita Vishwa Vidyapeetham, India*

Bharat Jayaraman and Venu Govindaraju, *State University of New York at Buffalo*

The ability to identify people and answer questions about their whereabouts in a cyberphysical space is critical to many applications. Integrating recognition with spatiotemporal reasoning enhances the overall performance of information retrieval.

A *cyberphysical space* is embedded with intelligence, providing a natural interface with humans using vision, speech, gestures, and touch, rather than keyboard and mouse. Key to realizing this paradigm is identifying and tracking people in the space. Indeed, the ability to identify and track people and answer questions about their whereabouts is critical to many applications.¹

The scenarios range from environments in which most of the individuals are known or preregistered, as in health-care monitoring, to those in which most of the individuals are unknown, as in homeland security. Consider two real-life scenarios:

- An elderly resident in an assisted living facility wears a radio-frequency identification (RFID) badge for continuous monitoring of his presence. On one occasion, he enters the elevator alone and is trapped due to a power failure. The RFID signals that his badge transmits aren't in the range of any receiver. Only much later, when the elevator resumes its service, do the attendants discover him.
- An intruder has gained illegal entry into a secure facility that surveillance cameras monitor. After an alert

sounds, security personnel set out to find the intruder. The search team relies on input from the control room personnel monitoring the facility through multiple video feeds. The intruder no longer appears on any of the video feeds. The search team has no other recourse but to search each room.

These scenarios illustrate the need for automated approaches to transforming multimedia data into a form suitable for information retrieval, a challenge that spans video and audio processing, computer vision, spatiotemporal reasoning, and data models. The scenarios also highlight the need for unobtrusive data gathering; people should be able to go about their normal activities without being subject to a pause-and-declare routine or the burden of RFID tags or badges.² Identifying people from their face, gait, or voice is more natural and less obtrusive and hence more suited for a cyberphysical space.

We postulate three fundamental operations for a cyberphysical space: *recognition*, *reasoning*, and *retrieval*—the three Rs. Biometric recognition based on modalities such as face, gait, and voice is inherently inexact and error prone; thus, a probability distribution typically represents the output of such a recognizer.³ Information retrieval in a

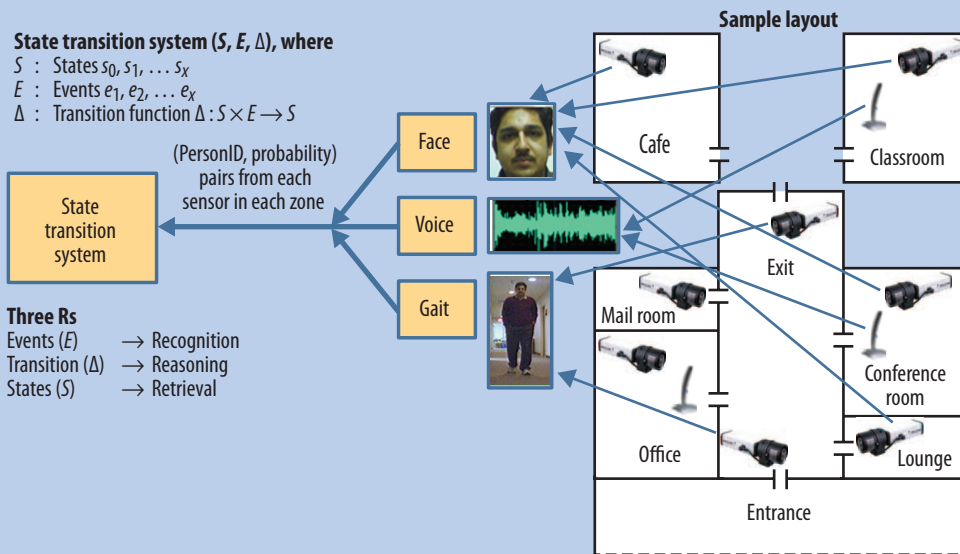


Figure 1. Using a state transition system to identify and track people in cyberphysical spaces. The system comprises states, events, and a transition function. The events abstract approaches to recognition; the transition function abstracts approaches to reasoning; and the states provide a basis for defining data models to support information retrieval.

cyberphysical space is concerned with the location of people at various points in time. Consequently, the queries will be probabilistic and spatiotemporal—for example, where was X last seen? or what's the probability that Y and Z met in the high-security zone between 6 p.m. and 7 p.m.? To alleviate the shortcomings of a purely recognition-based approach, we show that integrating recognition with spatiotemporal reasoning enhances the overall performance of retrieval in a cyberphysical space.

The extensive literature on ubiquitous or pervasive computing⁴ abounds in applications⁵ in which a physical device's actuation is at the paradigm's core. Actuation-based cyberphysical spaces tend to use simpler sensors, such as for temperature, pressure, and motion, and are more concerned with networking sensors and computing devices. In contrast, in our novel paradigm, information retrieval based on spatiotemporal queries is the main driver for identification and tracking.

A SCALABLE, UNIFIED APPROACH

Unlike earlier approaches to the problem of tracking people,⁶ our paradigm doesn't require massive numbers of sensing devices to cover the monitored space completely; rather, in a more realistic scenario, biometric devices embedded in only the main zones—such as hallways, entrances, and exits—capture data from a distance.⁷ This is a significant departure from current methods that tag people based on cues such as clothing or height and establish correspondences between adjacent fields of view interspersed by blind spots. Such approaches don't scale. Because we focus on indoor environments such as homes and offices,

which don't suffer the power or battery-life problems of outdoor environments, we can deploy and maintain sensors and other infrastructure with greater ease.

As Figure 1 shows, our approach provides a unified treatment of the three Rs using a novel state transition system that comprises states, events, and a transition function. This system's strength is that it accommodates different approaches to recognition, reasoning, and retrieval: the events abstract approaches to recognition; the transition function abstracts approaches to reasoning; and the states provide a basis for defining data models to support information retrieval.

This article extends our previous work, which focused on a purely recognition-based approach.⁸ Our state transition system is fundamentally probabilistic because the biometric recognition that underlies events is inexact. To evaluate the performance of identification and tracking in a cyberphysical space, we also formulate quantitative metrics based on two information-theoretical concepts: *precision* and *recall*. These concepts are standard performance measures in the information-retrieval literature, but we adapt their definitions to suit our context.

ABSTRACT MODEL FOR CYBERPHYSICAL SPACES

We abstract a cyberphysical space's behavior as a state transition system (S, E, Δ), where S is the set of states labeled $s_0, s_1, \dots, s_x$; E is the set of events labeled $e_1, e_2, \dots, e_x$; and $\Delta : S \times E \rightarrow S$ is a function that models the state transition when an event occurs.⁸ We can depict the state transitions as follows:

$$s_0 \xrightarrow{e_1} s_1 \xrightarrow{e_2} s_2 \dots \xrightarrow{e_x} s_x.$$

A *state* records for each zone and each occupant o_i , $i = 1 \dots n$, the probability of presence in that zone, $p_s(o_i)$. For each occupant, the sum of probabilities across all zones equals 1. The states abstract the information necessary to support information retrieval.

An *event* abstracts a biometric recognition step and is represented as a set of person-probability pairs, $\langle o_i, p(o_i) \rangle$, where $p(o_i)$ is the probability that the system recognized occupant o_i at this event. We also have

$$\sum_{i=1}^n p(o_i) = 1.$$

The *transition function* abstracts the reasoning necessary to affect state transitions. In the zone of occurrence, we define $p_s(o_i) = p(o_i) + x_i * p'_s(o_i)$, where $x_i = 1 - p(o_i)$ and $p'_s(o_i)$ is the probability of the occupant in the previous state. For all other zones, we define $p_s(o_i) = x_i * p'_s(o_i)$. This ensures that the sum of probabilities for an occupant across all zones in the resultant state equals 1.⁷

Events are assumed to be independent, but the transition function captures the dependency on the previous state, as in a Markov process. The “Hidden Markov Models” sidebar explains the difference between HMMs and our state transition system.

Table 1 illustrates this dependency, showing a sample state transition in a hypothetical four-zone cyberphysical space with five occupants. Event e_{11} occurs at zone 2 (z_2) and corresponds to the movement of occupant o_5 from z_1 to z_2 . The states s_{10} and s_{11} reflect the probability of the five occupants’ presence in each of the four zones before and after event e_{11} . The occupants listed in the last row correspond to the ground truth (G).

Because we don’t continuously monitor the environment, we record a discrete set of biometric recognition events corresponding to an occupant’s movements from one zone of the environment to another. The choice of biometric sensors for a zone can vary and depends on various factors. For example, face recognition might not be suitable in some zones for privacy

HIDDEN MARKOV MODELS

Hidden Markov models (HMMs) and their variants, such as factorial HMMs and coupled HMMs, are examples of dynamic Bayesian networks.¹ In these models, transition probabilities derive from empirical data gathered about people’s movements through the space over a period of time. Because we can’t assume a predictable pattern of movement of people through various cyberphysical space zones, we don’t adopt this approach.

In our state transition system, biometric capture devices provide direct information on event occurrences in specific zones. Given any event in a zone, the next state is unambiguously determined, although the state information is probabilistic. Furthermore, a state with n occupants and m zones would require only $m \times n$ storage, because for each of the m zones we record the probabilities of each of the n occupants being present in that zone.

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reasons, and voice recognition might not work well in a noisy zone.

Biometric recognition based on a single modality can be error prone, so fusing multiple modalities can improve the recognition process’s overall accuracy. When recognition is based on more than one biometric modality, the system fuses together the outputs from the individual recognizers to derive a single set of person-probability pairs. To detect an outsider’s presence, when a person bearing little or no resemblance to any of the registered occupants arrives at the entry zone, the biometric recognition step would produce low probabilities for all occupants.

EXPERIMENTAL TESTBED AND PERFORMANCE METRICS

We use an experimental testbed to validate our abstract model. Figure 2a shows the overall architecture, and Figure 2b shows the main interface of our simulator for a cyberphysical space. In a simulation run, a script randomly generates a set of trajectories for a given set of occupants. Each trajectory corresponds to one occupant’s movements

Table 1. Sample state transition.

Occupants	State s_{10}				Event e_{11}	State s_{11}			
	Zone 1 (z_1)	Zone 2 (z_2)	Zone 3 (z_3)	Zone 4 (z_4)	Zone 2 (z_2)	Zone 1 (z_1)	Zone 2 (z_2)	Zone 3 (z_3)	Zone 4 (z_4)
o_1	0.06	0.19	0.62	0.13	0.06	0.06	0.24	0.58	0.12
o_2	0.05	0.10	0.34	0.51	0.01	0.05	0.11	0.34	0.50
o_3	0.10	0.50	0.25	0.15	0.18	0.08	0.59	0.21	0.12
o_4	0.04	0.09	0.22	0.65	0.20	0.04	0.27	0.17	0.52
o_5	0.52	0.09	0.19	0.20	0.55	0.23	0.60	0.08	0.09
G	o_5	o_3	o_1	o_2, o_4	o_5	—	o_3, o_5	o_1	o_2, o_4

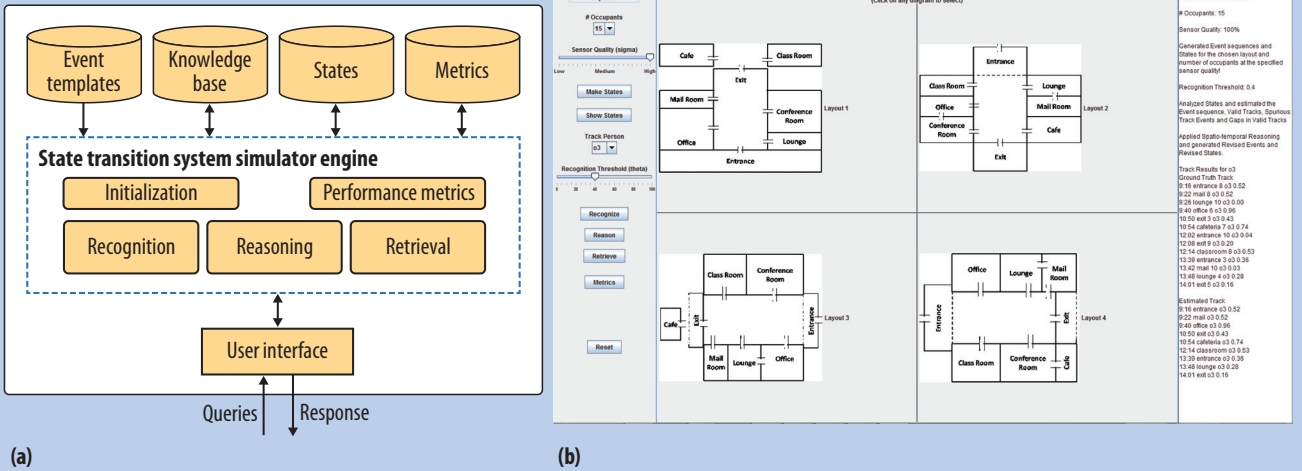


Figure 2. Simulator for cyberphysical spaces. (a) Overall architecture, including the state transition system simulator engine and the event templates, knowledge base, states, and metrics. (b) User interface.

and comprises a totally ordered sequence of events (where an event consists of a zone an occupant visited at a particular time) and the probabilities the biometric recognizer generated for this event. The user can choose a layout from a set of available options, specify the number of occupants in the space, and adjust parameters such as sensor quality (σ) and recognition threshold (θ) before a simulation run. The simulator also generates performance metrics associated with recognition before and after spatiotemporal reasoning.

We derived the simulation data using the face images of 45 individuals and three video cameras of varying image quality. We customized an OpenCV (www.intel.com/technology/computing/opencv/index.htm) implementation of the eigenface algorithm for this purpose. The video camera quality, as well as variations in pose, illumination, and expression, can cause fluctuations in the recognition process's overall accuracy, especially in unconstrained settings. Our formulation of sensor quality σ abstracts intrinsic and extrinsic factors that can affect recognition output. We used 10 different event templates (face images) for every individual. When the sensor quality is reduced (using the slider bar), the system chooses a lower-quality image such that the event probability for the person recognized is correspondingly lower. A varying number of false positives across these event templates accounts for the variability, typical of unconstrained biometric recognition.

To evaluate a cyberphysical space's performance, we define the concepts of precision (π) and recall (p) for a cyberphysical space in terms of the ground truth G :

$$\pi = t_p / (t_p + f_p), \text{ where } t_p \text{ is the set of true positives and } f_p \text{ is the set of false positives. The set } t_p = \{o_i : p_s(o_i) \geq \theta \wedge o_i \in \text{occ}(G)\}, \text{ while the set } (t_p + f_p) = \{o_i : p_s(o_i) \geq \theta\}.$$

$$p = t_p / (t_p + f_n), \text{ where } t_p \text{ is defined as above, and } f_n \text{ is the set of false negatives. The set } (t_p + f_n) = \{o_i : o_i \in \text{occ}(G)\}.$$

For a given input event sequence, ground truth G is a sequence of states wherein the presence or absence of any occupant in any zone is known with certainty (0 or 1). Precision captures the extent of false positives, and recall captures the extent of false negatives. These definitions are stated in terms of a recognition threshold θ ; only those people with a probability $\geq \theta$ are assumed to be present. When a person's probability in two or more zones is $\geq \theta$, the zone with the highest probability is taken as the zone of that person's presence. We refer to the set of people occurring in a ground truth G as $\text{occ}(G)$.

We plotted performance metrics across multiple runs for 15 occupants. Figure 3a plots average precision and average recall for varying values of recognition threshold θ at sensor quality $\sigma = 1.0$. Note that the average precision increases up to $\theta = 0.6$ and then declines. The average precision is low at low values of θ , because a high proportion of false positives is present in the set of recognized occupants. As θ increases, the proportion of false positives diminishes until reaching a point of inflexion. From this point, the average precision begins to decline because the true positives also fail to get recognized. Average recall decreases with increasing θ , because the proportion of false negatives steadily increases with θ .

Figures 3b and 3c show the average precision and average recall curves for varying sensor quality σ . The dependence of precision and recall on the recognition threshold θ causes the precision curves in Figure 3b to assume a bell shape. Depending on the application, the

user can choose the recognition threshold for a cyberphysical space. For example, in security-related applications, the user might minimize the number of false positives, whereas in assisted living scenarios, the user might want to minimize the false negatives. In the absence of any additional information, a reasonable operating point is one in which false positives equal false negatives. At any given σ , we can obtain such a recognition threshold θ from the intersection of the average precision and average recall curves, as Figure 3a shows.

SPATIOTEMPORAL REASONING

Our experimental results for a purely recognition-based approach (the blue curve in Figure 4) show that as the number of people in the cyberphysical space increases, so does the number of spuriously identified people. To minimize recognition errors' impact, the key is to determine a person's identity on the basis of information from a track of events and corresponding states rather than a single event. The reason is that consecutive track elements of a valid track will mostly obey the zone adjacencies in the physical environment, whereas spurious tracks will mostly violate the zone adjacencies. A transition function of the form $\Delta : P(S) \times E \rightarrow S$ can capture this track-based reasoning. The transition function takes as input a set of previous states, computes the occupant tracks from these states, and determines the next state when an event occurs.

To determine the tracks from a set of states, the transition function determines for each event occurrence the individual who moved between two zones of the cyberphysical space. From this information, the system immediately obtains the set of all tracks (both valid and spurious). The criterion for determining which occupant moved is defined in terms of the maximum difference in occupant probabilities between two consecutive states in the event's zone of occurrence. It might appear that the person with the highest probability in an event is the one who moved. However, we don't adopt this approach for two reasons: event information could be erroneous; and comparing consecutive states gives due importance to both event and historical information, which the system captures and retains only in the states.

The initial states of a cyberphysical space are likely to have more errors because track-based

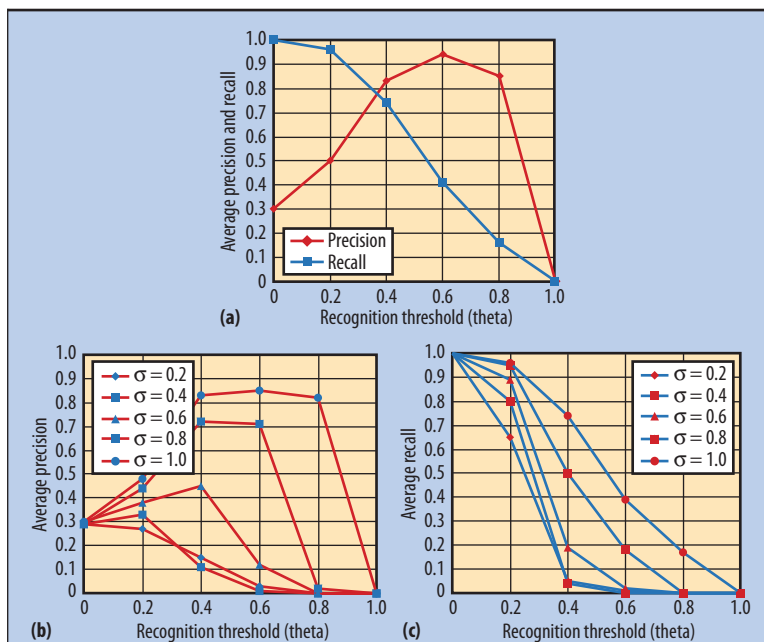


Figure 3. Precision and recall metrics. (a) Average precision and average recall versus theta shows that average precision increases up to $\theta = 0.6$ and then declines. (b) In average precision curves for varying sigma, the dependence of precision on the recognition threshold θ results in the bell-shaped precision curves. (c) Average recall curves for varying sigma show that average recall decreases with increasing θ .

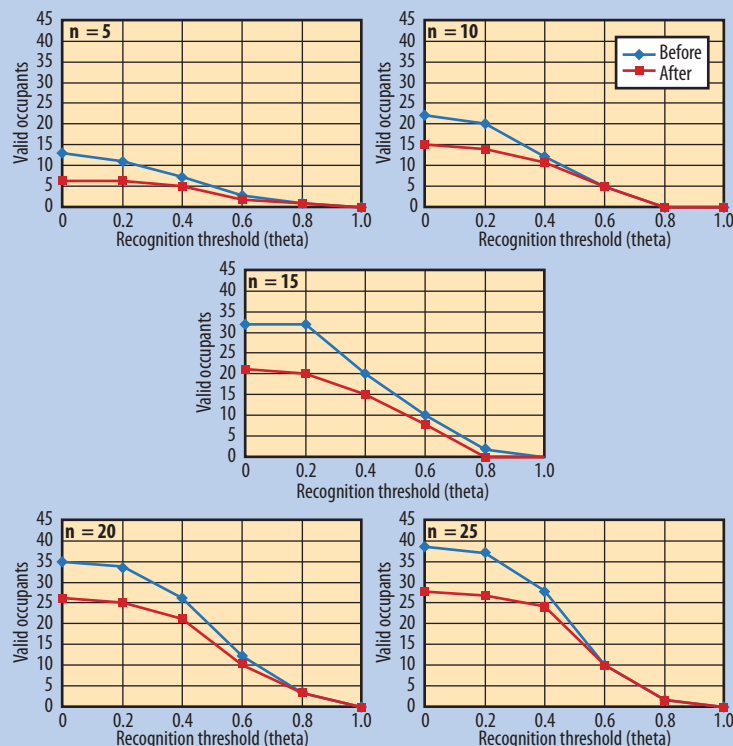


Figure 4. Estimated number of occupants before and after reasoning. The benefits of reasoning are more pronounced at a lower value of θ , in which the number of false positives is higher.

RELATED WORK ON SPATIOTEMPORAL REASONING

Spatiotemporal reasoning over occupant tracks is similar to a higher-order Markov process, because the next state depends on multiple previous states. When the transition function also updates the information in previous states, the resulting inference is closer to that of a Markov Random Field (MRF) analysis.

In the MRF approach, we can model the operation of a cyber-physical space with an undirected graph whose nodes correspond to space-time (or zone-event) points and edges capture space-time adjacency. Spatiotemporal reasoning with MRF is based on a neighborhood analysis around an event's zone of occurrence. Although it's more general in principle, it's also computationally more complex than track-based reasoning, which is more specialized and hence can more efficiently incorporate a global view of the system. Researchers have also investigated spatiotemporal reasoning from a logic and constraint perspective, with applications in geographical information systems, computer vision, planning, and so on.¹

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RELATED WORK ON SPATIOTEMPORAL DATABASES

The data in cyberphysical spaces is fundamentally probabilistic and spatiotemporal because people are moving between zones over a period of time, and we're interested in their trajectories. Hence, the data models and query languages of interest to us in a cyberphysical space are also probabilistic and spatiotemporal.

Considerable research has focused on spatiotemporal databases during the past two decades. For example, location-based systems have been a major driver for the interest in moving object databases and their associated data models, query languages, indexing, and uncertainty.^{1,2}

In addition to the challenges involved in spatiotemporal databases, research into probabilistic databases has gained momentum over the years. This growth is owing to the emergence of a broad range of applications that must manage large and imprecise data sets in domains such as sensor networks, information extraction, and business intelligence.³ Our research in cyberphysical spaces makes crucial use of probabilistic and temporal concepts, while we treat the spatial issues more qualitatively (symbolic) than quantitatively (geometric).

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reasoning on shorter tracks is less effective in mitigating the errors owing to recognition. Over a period of time, as longer tracks form, the reasoning process not only determines subsequent states with less error, it can also correct the errors in the initial states. Such a transition function would have the form $\Delta : P(S) \times E \rightarrow P(S)$. That is, it takes a set of states as input, computes the tracks from these states, and determines as output the next state along with a revised set of previous states.

Figure 4 shows the benefits of integrating recognition and track-based reasoning (indicated by the red curve) to reduce the extent of spuriously identified occupants. The benefits of reasoning are more pronounced at a lower value of θ , in which the number of false positives is higher. The "Related Work on Spatiotemporal Reasoning" sidebar provides more details.

INFORMATION RETRIEVAL

The state transition system provides a natural basis for retrieving answers to queries about occupants' whereabouts in the cyberphysical space. We show how to formulate spatiotemporal queries using the SQL database query language, focusing on computing probabilities, an aspect novel to our model. For more information, see the "Related Work on Spatiotemporal Databases" sidebar.

We also have developed more complex queries in a constraint-based extension of logic programs, called constraint logic programming (Real), or CLP(R). This extension permits general recursive queries and reasoning over real-valued variables and arithmetic operations.⁹

We define an occupancy relation, *occupancy(start, end, person, zone, probability)*, where *start* and *end* define a time interval comprising a discrete totally ordered set of points (because events are also discrete). The attribute *probability* $\in \mathbb{R}$, the set of real numbers, is functionally dependent on the other four attributes. This relation captures the state information after the system has performed recognition and reasoning and determined a set of valid occupants. SQL queries' basic syntax is as follows: *SELECT attributes FROM relations WHERE condition*. The *condition* is typically a conjunction of simpler tests that serve as a basis for tuple selection. This basic syntax has numerous extensions for performing aggregate operations, grouping, ordering, and so on.

Consider the following query: What is the probability that occupant 7 was in the lounge at any time between 10 a.m. and 11 a.m.? Because multiple subintervals can exist from 10 a.m. to 11 a.m., while o_7 was in the lounge (with different probabilities), the answer to the query is 1 minus the product of the probabilities that o_7 wasn't in the lounge during every such subinterval:

```
(1 - PROD(
  SELECT (1-probability) as prob2
  FROM occupancy
  WHERE person = o7 and
    zone = lounge and
    10:00 <= start and end <= 11:00)
)
```

The probability that an occupant wasn't in the lounge at a given time is 1 minus the probability that he was in the lounge at this time, because the sum of the probabilities across all zones equals 1 at any given time.

Although query-independent performance characterization is holistic at a system level, characterizing a cyberphysical space's performance with respect to queries might be more meaningful and better cater to a user's interest. The performance metrics of any given query are defined in terms of the ground truth, which is a set of true answers associated with the query. The nature of the response set might vary depending on the type of query the user poses. It might also comprise entities such as occupants and zones, or attributes such as probabilities of presence and time of occurrence. The response set can likewise include derived attributes such as duration of presence, tracks, and so on, on the basis of relations defined as part of the data model. From an information-retrieval perspective, precision is the fraction of retrieved answers relevant to the query, and recall is the fraction of the answers relevant to the query that the system successfully retrieved.

Our state transition model serves as an effective basis for developing practical deployments; in particular, the precision recall curves help determine a suitable operating point for fine-tuning the cyberphysical space to suit the application at hand. Our current focus has been on relatively small indoor spaces, such as offices and nursing homes, where it is possible to know registered occupants in advance. We plan to explore larger indoor spaces, such as airport terminals, where our interest is in intruder identification and tracking. Such spaces also require us to address the system issues that arise out of networking a large number of cameras. 

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Vivek Menon is an assistant professor of information systems in the School of Business at Amrita Vishwa Vidyapeetham (Amrita University), India. His research interests include intelligent systems and smart environments. Menon received a PhD in computer science from Amrita University. He is a member of IEEE, the ACM, and the Association for Information Systems. Contact him at vivek_menon@cb.amrita.edu.

Bharat Jayaraman is a professor in the Computer Science and Engineering Department at the State University of New York at Buffalo. His research interests include software systems and languages. Jayaraman received a PhD in computer science from the University of Utah. He is a senior member of IEEE and a member of the ACM. Contact him at bharat@buffalo.edu.

Venu Govindaraju is a Distinguished Professor and directs the Center for Unified Biometrics and Sensors at SUNY Buffalo. His research interests include pattern recognition applied to biometrics and document analysis. Govindaraju received a PhD in computer science from SUNY Buffalo. He's a Fellow of IEEE, the ACM, the American Association for the Advancement of Science, and the International Association for Pattern Recognition. Contact him at govind@buffalo.edu.

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